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STORAGE DEVICE, STORAGE SYSTEM AND OPERATING METHOD OF THE SAME USING MEMORY BUFFER

Abstract

A storage device, a storage system and an operating method of the storage device are provided. The storage device connected to a host comprises a storage controller including an address translation cache that translates a virtual address received with a command from the host into a physical address, wherein the storage controller stores index information on a host specific memory buffer (SMB) in the host in advance, performs direct memory access to the virtual address for the host SMB when a cache miss occurs in the address translation cache, and transmits an exceptional completion queue (CQ) including an indicator to the host when the direct memory access is completely performed.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority from Korean Patent Application No. 10-2024-003155 filed on Mar. 5, 2024, in the Korean Intellectual Property Office and all the benefits accruing therefrom under 35 U.S.C. 119, the contents of which in its entirety are herein incorporated by reference.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a storage device.

Description of the Related Art

[0003] A storage system may provide address translation for direct memory access (DMA) from input/output devices (e.g., network adapters, graphic process units (GPUs), storage controllers, and the like) to a main memory of the storage system. The storage system may include an input/output memory management unit (IOMMU) for protecting a memory from an error operation due to a limited size of the main memory that may be accessed by the input/output devices.

[0004] The storage system may include an input/output translation lookaside buffer (IOTLB) to improve performance of the input/output memory management unit. The input/output translation lookaside buffer may be used as a cache for increasing an address verification speed. However, a cache miss of the input/output translation lookaside buffer may be a relatively greater factor in performance degradation of the storage system than an untranslated addressing system.

[0005] To mitigate the cache miss of the input/output translation lookaside buffer, the storage system may include a device-side input/output translation lookaside buffer (or referred to as a translation lookaside buffer (TLB) or an address translation cache (ATC); referred to as ATC for differentiation from a host side) in a device-side (e.g., host interface in a storage device) thereof. The ATC may support address translation services (ATS) defined by Peripheral Component Interconnect-Special Interest Group (PCI-SIG®) and/or Peripheral Component Interconnect Express (PCIe®).

[0006] As the device-side of the storage system performs the address translation service, the ATC may be in charge of at least a portion of address processing processed by a central processing unit (CPU) and/or the IOMMU, and may increase the overall size of the input/output translation lookaside buffer to a size of a sum of the input/output translation lookaside buffer and the ATC.

[0007] Since an access size (e.g., minimum data size to be read) to a non-volatile memory device in the storage system and an access size of a memory in the host system are different from each other, performance of the storage system may be improved when data is read based on the non-volatile memory device.

[0008] However, when data is read based on the non-volatile memory device, the number of virtual addresses to be translated by the ATC may be increased, and thus an address translation cache miss rate (ATC miss rate) may be increased. Due to the limited size of the ATC, the storage system may suffer performance degradation caused by ATC Miss, and ATC update latency performed during the ATC Miss becomes variable depending on the configuration of the storage system, making it difficult to predict.

BRIEF SUMMARY

[0009] According to an aspect of the present disclosure, there is provided a storage device comprises a storage controller including an address translation cache that translates a virtual

address received with a command from a host into a physical address, wherein the storage controller stores index information on a host specific memory buffer (SMB) in the host in advance, performs direct memory access to the virtual address for the host SMB when a cache miss occurs in the address translation cache, and transmits an exceptional completion queue (CQ) including an indicator to the host when the direct memory access is completely performed.
[0010] According to another aspect of the present disclosure, there is provided a storage system comprises a host including a host memory storing index information on a physical address and a host processor outputting a command and a virtual address, and a storage device sharing the index information in advance and transmitting data according to the command to the host memory together with an indicator according to the index information in accordance with a direct memory access operation.
[0011] According to another aspect of the present disclosure, there is provided a method for operating a storage device, comprises performing direct memory access to a host together with a virtual address when a cache miss for the virtual address occurs in an address translation cache, and outputs a completion queue, which includes an indicator corresponding to the virtual address, to the host.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0012] The above and other aspects and features of the present disclosure will become more apparent by describing in detail exemplary embodiments thereof with reference to the attached drawings, in which:
[0013] FIG. **1** is a block diagram illustrating a storage system according to some embodiments;
[0014] FIG. **2** is a block diagram illustrating a host memory according to some embodiments;
[0015] FIG. **3** is a table illustrating index information shared in advance by a storage device and a host in accordance with some embodiments;
[0016] FIG. **4** is a flow chart illustrating an operating method of a storage system;
[0017] FIG. **5** is a flow chart illustrating an operating method of a storage system according to some embodiments;
[0018] FIG. **6** is a block diagram illustrating the operating method of FIG. **5**;
[0019] FIG. **7** is a flow chart illustrating an operating method of a storage system according to some embodiments;
[0020] FIGS. **8** and **9** are flow charts illustrating an operating method of a storage system according to some embodiments;
[0021] FIG. **10** is an exemplary block diagram illustrating a storage system to which a storage device according to some embodiments is applied; and
[0022] FIG. **11** is an exemplary block diagram illustrating a data center to which a storage device according to some embodiments is applied.
DETAILED DESCRIPTION OF THE DISCLOSURE
[0023] FIG. **1** is a block diagram illustrating a storage system according to some embodiments, and FIG. **2** is a block diagram illustrating a host memory according to some embodiments. FIG. **3** is a table illustrating index information shared in advance by a storage device and a host in accordance with some embodiments.
[0024] Referring to FIGS. **1** to **3**, a storage system according to some embodiments may be, for example, a mobile system such as a mobile phone, a smart phone, a tablet personal computer (PC), a wearable device, a healthcare device or an Internet of things (IoT) device. However, a storage system **1** of FIG. **1** is not necessarily limited to a mobile system, but may be a personal computer, a laptop computer, a server, a media player or an automotive device such as a navigator.

[0025] Subsequently, referring to FIG. **1**, the storage system **1** according to some embodiments includes a host **100** and a storage device **2**.
[0026] The host **100** includes a host controller **110** (also referred to herein as host processor), a host memory **130** and an input/output memory management unit (IOMMU) **120**. The IOMMU **120** includes, for example, a translation agent (TA) **121**, and an address translation and protection table (ATPT) **122**.
[0027] The host controller **110** may manage an operation of storing data (e.g., recorded data) of a data buffer **131** in the host memory **130** in the storage device **2** or storing data (e.g., read data) of the storage device **2** in the data buffer **131**.
[0028] The translation agent **121** may include hardware, firmware and/or software that translates an address in a PCIe transaction into a physical address associated therewith. For example, when receiving an address translation service request from the storage device **2**, the translation agent **121** allocates a new physical address to a virtual address included in the address translation service request and replies the allocated physical address to the storage device **2200**. The physical address may be a location in the host memory **130**.
[0029] The address translation and protection table **122** may store address translation information processed by the translation agent **121** to process PCIe requests such as direct memory access (DMA) read or direct memory access write among address translation services (ATS).
[0030] The host memory **130** may be an embedded memory provided in the host **100**, or may be a non-volatile memory or a memory module, which is disposed outside the host **100**. The host memory **130** may include, for example, a data buffer **131**, a host specific memory buffer (SMB) **132** and a host completion manager **133**.
[0031] The data buffer **131** may serve as a buffer memory for temporarily storing data to be transmitted to the storage device **2** or data transmitted from the storage device **2**.
[0032] The host SMB **132** stores index information, and also stores data requested for direct memory access only with a virtual address without a physical address from the storage device **2**. The host completion manager **133** receives and stores a normal completion queue having no indicator or an exceptional completion queue including an indicator from the storage device **2**. The host completion manager **133** checks (or extracts) the indicator from the exceptional completion queue.
[0033] The translation agent **121** maps the virtual address received together with the direct memory access request to a physical address of an index corresponding to the checked indicator and stores address translation information between the virtual address and the mapped physical address in the ATPT **122**.
[0034] According to one embodiment, the host controller **110** and the host memory **130** may be implemented as separate semiconductor chips. Alternatively, in some embodiments, the host controller **110** and the host memory **130** may be integrated into the same semiconductor chip. As an example, the host controller **110** may be any one of a plurality of modules provided in an application processor, and the application processor may be implemented as a system on chip (SoC).
[0035] The storage device **2** may further include a storage controller **200** and a non-volatile memory **300**. The storage controller **200** may control data writing and reading operations for the non-volatile memory by executing a flash translation layer.
[0036] The storage controller **200** may include an address translation cache (ATC) **210**, a direct memory access manager **220**, a device completion manager (also referred to herein as NVMe completion manager) **230** and a storage memory buffer (SMB) **400**.
[0037] The address translation cache **210** is a cache for increasing an address verification speed, and stores a physical address of the host memory **130**, which corresponds to a virtual address received together with a command by the host processor. The address translation cache **210** may support address translation services (ATS) defined by a peripheral component inter-connect-special

interest group (PCI-SIG®) and/or a peripheral component interconnect express (PCIe®). When there is no information on a physical address corresponding to a virtual address in the address translation cache **210** (Cache Miss), the storage controller **200** transmits an address translation service request (ATS request) to the host **100**. The address translation service request (ATS Req) may be associated with one or more address translation cache entry replacement policies (e.g., user definition, QoS, rate limit or workload-based policy). Alternatively, the address translation service request (ATS Req) may be associated with an address translation cache (ATC) replacement algorithm (e.g., Deficit Weighted Round Robin (DWRR).
[0038] The host **100** may transmit an address translation service response (ATS Resp) to the address translation service request (ATS Req) to the storage device **2**. The address translation service response (ATS Resp) may be a completion response to a request from the storage device **2**, or may be a cache miss response that physical address information requested by the storage device **2** does not exist.
[0039] The direct memory access manager **220** is coupled to an internal communication bus of the storage device **2**, and transfers data between the host **100** and the non-volatile memory **300**. The direct memory access manager **220** may perform a write operation by allowing data received from the data buffer **131** of the host **100** to directly access the non-volatile memory **300**, or may transmit data read from the non-volatile memory **300** to the data buffer **131** of the host **100** or the host SMB **132** by direct memory access.
[0040] The storage memory buffer (SMB) **400** may include an SMB manager **410** and a device SMB **420**. When a cache miss occurs in the address translation cache **210**, the SMB manager **410** requests the direct memory access manager **220** of memory access to the virtual address. In addition, the SMB manager **410** selects any one indicator from index information stored in advance with respect to the virtual address for which the direct memory access is requested and transmits the index information to the device completion manager **230** together with the selected indicator to perform a completion notification. For example, the index information may be stored in the device SMB (N.SMB) **420** in advance. The device SMB (N.SMB) **420** may be implemented as, for example, a static random access memory.
[0041] When the direct memory access manager **220** completely performs the direct memory access, the device completion manager (N.Completion) **230** transmits a completion queue (CQ) to the host completion manager **133**. The completion queue may be, for example, an exceptional completion queue including an indicator, or may be a normal completion queue having no indicator. When the device completion manager **230** receives a completion notification together with the selected indicator from the SMB manager **410**, the device completion manager **230** transmits the exceptional completion queue to the host completion manager **133**. However, when the direct memory access manager **220** performs the direct memory access with the physical address, the device completion manager **230** transmits the normal completion queue to the host completion manager **133**.
[0042] The storage device **2** may include storage media for storing data in accordance with a request from the host **100**. The storage device **2** may be a device that complies with the non-volatile memory express (NVMe) standard. The host **100** and the storage device **2** may generate and transmit packets according to standard protocols, which are employed, respectively.
[0043] The storage device **2** may include a non-volatile memory **300**. When the non-volatile memory includes a flash memory, the flash memory may include a two-dimensional NAND memory array or a three-dimensional (or vertical) NAND memory array. As another example, the storage device **2** may include various other types of non-volatile memories **300**. For example, the storage device **2** may include a magnetic RAM (MRAM), a spin-transfer torque MRAM, a conductive bridging RAM (CBRAM), a Ferroelectric RAM (FeRAM), a phase RAM (PRAM), a resistive RAM and other various types of memories.
[0044] Referring to FIG. **3**, the host **100** and the storage device **2** may share index information in

advance before an access operation. For example, when the system is booted, the storage device **2** may receive and store the index information from the host **100** and then perform an access operation according to a command from the host **100**.
[0045] The index information is information on the host memory **130**, and includes information on a storage location in the host SMB **132**. The index information may include, for example, a physical address, a block size, and an index number corresponding to an indicator.
[0046] For example, when the storage device **2** performs direct memory access to the host memory **130** only with a virtual address without a physical address, the host **100** controls the storage device **2** to perform the direct memory access to the host SMB **132**. In this case, the host checks the indicator included in the completion queue, and processes the corresponding direct memory access from the physical address in the host SMB **132** of the index corresponding to the indicator.
[0047] FIG. **4** is a flow chart illustrating an operating method of a storage system.
[0048] Referring to FIG. **4**, the host **100** transmits a data write command SQ to the storage controller **200** in a command (CMD) phase. The data write command SQ may be, for example, transferring a command (e.g., a write command) through a submission queue when the host **100** transfers a signal to the storage device **2**.
[0049] When receiving a data write command, the storage device **2** translates a virtual address included in the data write command SQ into a physical address in an address translation (PTR) phase. In more detail, the storage controller **200** requests whether there is physical address information corresponding to the virtual address received in the address translation cache **210** (VA-PA Query), and when a cache miss occurs in the address translation cache **210** (ATC Miss), the address translation cache **210** transmits an address translation service request (ATS REQ) including the cache missed virtual address to the translation agent **121** of the host **100**.
[0050] The translation agent **121** checks the physical address mapped to the received virtual address, and replies the address translation service response (ATS RSP) to the address translation cache **210** through the storage controller **200** when there is no mapped physical address in the ATPT **122**. The address translation cache **210** transmits a page request for requesting allocation of a new physical address, that is, a page, to the virtual address, to the translation agent **121**, and the translation agent **121** allocates a new page based on the ATPT **122** and then performs a page request success response. Afterwards, the address translation cache **210** retransmits the address translation service request (ATS REQ), which includes the cache missed virtual address, to the translation agent **121** of the host **100**, and the translation agent **121** performs the address translation service response (ATS RSP, Page Hit) that includes a physical address of the newly allocated page. The storage controller **200** extracts the newly received physical address from the address translation service response and transfers the extracted physical address to the address translation cache **210**. The address translation cache **210** updates the newly received physical address with information on the corresponding virtual address (ATC Fill), and transmits the updated physical address to the direct memory access manager **220**.
[0051] In the direct memory access phase (DMA phase), the direct memory access manager **220** performs direct memory access to the host memory **130** of the updated physical address (DMA to PA). Subsequently, the device completion manager **230** transmits the completion queue CQ to the host **100**.
[0052] When the operation in the address translation phase continues to be repeated as shown in FIG. **4** whenever a cache miss occurs in the address translation cache **210**, latency is increased during the operation, which may lead to performance degradation of the storage system **1**.
[0053] To solve this problem, performance degradation of the storage system **1** may be mitigated when the latency in the address translation phase is reduced even though a cache miss occurs, as described below with reference to FIG. **5**.
[0054] FIG. **5** is a flow chart illustrating an operating method of a storage system **1** according to some embodiments, and FIG. **6** is a block diagram illustrating the operating method of FIG. **5**.

[0055] Referring to FIG. **5**, the host **100** shares index information on the host SMB **132** in the storage device **2** in advance. The storage device **2** may store the shared index information in the device SMB **420**, and the SMB manager **410** may use the stored index information during the operation. Afterwards, when receiving the command and the virtual address of the host **100**, the storage device **2** checks whether there is a physical address mapped to the virtual address in the address translation cache **210** (S**10**). When there is no mapped physical address in the address translation cache **210** and thus a cache miss occurs (ATC Miss, Cache Miss), the storage device **2** performs direct memory access to the host SMB **132** together with the virtual address VA (S**11**). Subsequently, the storage device **2** transmits an exceptional completion queue, which includes the indicator of the location where the corresponding direct memory access is performed, to the host **100** (S**12**). That is, since the storage device **2** performs the direct memory access to the host **100** together with the virtual address without address translation in the address translation phase of S**11** and performs the direct memory access at a physical address corresponding to the indicator in S**12**, the process performed from the cache miss (ATC Miss) to the cache update (ATC Fill) of FIG. **4** may be omitted, whereby latency according to the address translation operation may be reduced.
[0056] In more detail, referring to FIG. **6**, the host **100** shares the index information on the host SMB **132** in the storage device **2** in advance ({circle around (**1**)}). For example, the host SMB **132** and the device SMB **420** may store the same index information, respectively. The host processor **110** transmits a data write command (SQ Write) to the storage controller **200** ({circle around (**2**)}). The storage controller **200** checks the virtual address included in the host command from the ATC **210**. When a cache miss occurs because there is no physical address on the virtual address in the ATC **210** ({circle around (**3**)} ATC Miss), data related to the host command is stored in the device SMB **420**.
[0057] The SMB manager **410** makes a request to the direct memory access manager **220** for memory access to the virtual address ({circle around (**4**)} Request DMA). The direct memory access manager **220** performs the direct memory access to the host SMB **132** of the host **100** together with the virtual address based on the request of the SMB manager **410** ({circle around (**5**)} DMA to SMBh). The SMB manager **410** performs a completion request by notifying the device completion manager **230** of the location of the host SMB **132** directly accessed by the direct memory access manager **220** ({circle around (**6**)} Request Completion). The device completion manager **230** transmits the completion queue, which includes the indicator, to the host completion manager **133** ({circle around (**7**)} Transfer Completion).
[0058] The host **100** checks the indicator received by the host completion manager **133**, maps the same to the virtual address of the direct memory access performed in the host SMB **132** and notifies the translation agent **121** of the mapped result. The translation agent **121** updates address translation information by mapping a physical address having an index number corresponding to the indicator in the index information to the virtual address. For example, the updated address translation information may be stored in the ATPT **122**, and then, may be transferred upon request of the translation agent **121**.
[0059] FIG. **7** is a flow chart illustrating an operating method of a storage system according to some embodiments.
[0060] Referring to FIG. **7**, the command phase of the storage system **1** may be the same as that of FIG. **4**. However, when a cache miss occurs in the address translation phase (PTR phase) (ATC Miss), the storage controller **200** of the storage device **2** operates by itself and does not perform a communication operation with the host **100** in the SMB phase related to the operation in the storage memory buffer **400**. The SMB manager phase may be, for example, the steps {circle around (**2**)} and {circle around (**3**)} of FIG. **6**.
[0061] The storage controller **200** then performs the direct memory access to the virtual address in the DMA phase (DMA to SMBh) and transmits a completion queue including an indicator (CQ with SMB Indicator).

[0062] As described above, since the storage system **1** according to some embodiments transmits indicators while immediately performing memory access without a physical address based on index information without performing a process of asking and responding address translation information to the host in the address translation phase and SMB phase, latency due to address translation may be reduced.
[0063] FIGS. **8** and **9** are flow charts illustrating an operating method of a storage system according to some embodiments.
[0064] Referring to FIGS. **8** and **9**, the host **100** shares index information on the host SMB **132** in the storage device **2** in advance. Afterwards, when receiving the command and the virtual address of the host **100**, the storage device **2** searches whether there is a physical address mapped to the virtual address in the address translation cache **210** (S**20**, VA-PA Query).
[0065] When a cache miss occurs in the address translation cache **210** due to no mapped physical address (ATC Miss, Cache Miss), the storage device **2** transmits an address translation service request to the translation agent **121** of the host (S**21**, ATS REQ). The host transmits an address translation service response (ATS RSP) corresponding to the address translation service request to the storage controller **200**.
[0066] In the address translation service response, when there is no physical address mapped to the virtual address in the translation agent **121** (Page Miss), the SMB phase described with reference to FIG. **7** is performed. Then, the direct memory access to the host SMB **132** is performed together with the virtual address VA (S**22**, DMA to SMBh). Subsequently, the storage device **2** transmits a completion queue, which includes an indicator of a location where the corresponding direct memory access is performed for the completion queue, to the host **100** (S**23**, CQ with SMB Indicator).
[0067] When there is a physical address mapped to the virtual address in the translation agent **121** (Page Hit), an address translation service response including the mapped physical address is replied, and the address translation cache is updated in the address translation phase without the SMB phase (S**24**, DMA to PA). Afterwards, the storage controller **200** performs the direct memory access to the updated physical address (S**25**), and transmits a normal completion queue (Normal CQ) having no indicator to the host **100** (S**26**).
[0068] Unlike the embodiment of FIG. **7**, in the embodiment of FIG. **9**, the address translation service attempts to request the translation agent of the host at least once after the cache miss occurs, and the steps S**22** and S**23** corresponding to S**11** and S**12** of FIG. **5** are performed only when a page miss occurs in the translation agent. In this case, it is possible to reduce the latency of the request (Page Request) for allocating a page or the new address translation service response (ATS Rsp (Page Hit)), which has been described above with reference to FIG. **4**.
[0069] FIG. **10** is an exemplary block diagram illustrating a storage system to which a storage device according to some embodiments is applied.
[0070] FIG. **10** illustrates a system **1000** to which a storage device according to one embodiment is applied. The system **1000** of FIG. **10** may be a mobile system such as a mobile phone, a smart phone, a tablet personal computer (PC), a wearable device, a healthcare device or an Internet of things (IOT) device. However, the system **1000** of FIG. **10** is not necessarily limited to the mobile system, and may be a personal computer, a laptop computer, a server, a media player or an automotive device such as navigator.
[0071] Referring to FIG. **10**, the system **1000** may include a main processor **1100**, memories **1200***a* and **1200***b* and storage devices **1300***a* and **1300***b*, and may further include one or more of an image capturing device **1410**, a user input device **1420**, a sensor **1430**, a communication device **1440**, a display **1450**, a speaker **1460**, a power supplying device **1470** and a connecting interface **1480**.
[0072] Although not shown, the storage devices **1300***a* and **1300***b* may include the host interface described with reference to FIGS. **1** to **11**.
[0073] The main processor **1100** may control the overall operation of the system **1000**, more

specifically the operation of other elements constituting the system **1000**. The main processor **1100** may be implemented as a general purpose processor, a dedicated processor or an application processor.

[0074] The main processor **1100** may include one or more CPU cores **1110**, and may further include a controller **1120** for controlling the memories **1200***a* and **1200***b* and/or the storage devices **1300***a* and **1300***b*. In accordance with the embodiment, the main processor **1100** may further include an accelerator **1130** that is a dedicated circuit for high-speed data computation such as an artificial intelligence (AI) data computation. The accelerator **1130** may include a graphics processing unit (GPU), a neural processing unit (NPU) and/or a data processing unit (DPU), and may be implemented as a separate chip physically independent of other elements of the main processor **1100**.

[0075] The memories **1200***a* and **1200***b* may be used as main memory devices of the system **1000**, and may include a volatile memory such as an SRAM and/or a DRAM but may also include a non-volatile memory such as a flash memory, a PRAM and/or an RRAM. The memories **1200***a* and **1200***b* may be implemented in the same package as the main processor **1100**.

[0076] The storage devices **1300***a* and **1300***b* may serve as non-volatile storage devices for storing data regardless of whether a power source is supplied, and may have a storage capacity relatively greater than that of the memories **1200***a* and **1200***b*. In some embodiments, the storage devices **1300***a* and **1300***b* may include storage controllers **1310***a* and **1310***b* and non-volatile memories (NVM) **1320***a* and **1320***b* for storing data under the control of the storage controllers **1310***a* and **1310***b*. The non-volatile memories **1320***a* and **1320***b* may include a two-dimensional (2D) or three-dimensional (3D) vertical NAND (V-NAND) flash memory, but may include other types of non-volatile memories such as PRAM and/or RRAM.

[0077] The storage devices **1300***a* and **1300***b* may be included in the system **1000** in a physically separated state from the main processor **1100**, or may be implemented in the same package as the main processor **1100**. In addition, the storage devices **1300***a* and **1300***b* may be detachably coupled to other elements of the system **1000** through an interface, such as the connecting interface **1480**, which will be described later, by having the same form as that of a solid state device (SSD) or a memory card. The storage devices **1300***a* and **1300***b* may be devices that comply with a standard protocol such as Universal Flash Storage (UFS), embedded multi-media card (eMMC) or non-volatile memory express (NVMe), but is not necessarily limited thereto.

[0078] The image capturing device **1410** may capture a still image or a video, and may be a camera, a camcorder and/or a webcam.

[0079] The user input device **1420** may receive various types of data input from a user of the system **1000**, and may be a touch pad, a keypad, a keyboard, a mouse and/or a microphone.

[0080] The sensor **1430** may sense various types of physical quantities that may be acquired from the outside of the system **1000** and convert the sensed physical quantities into an electrical signal. The sensor **1430** may be a temperature sensor, a pressure sensor, an illuminance sensor, a position sensor, an acceleration sensor, a biosensor and/or a gyroscope sensor.

[0081] The communication device **1440** may perform transmission and reception of signals between other devices outside the system **1000** in accordance with various communication protocols. Such a communication device **1440** may be implemented by including an antenna, a transceiver and/or a modem.

[0082] The display **1450** and the speaker **1460** may serve as output devices that output visual information and auditory information to a user of the system **1000**, respectively.

[0083] The power supplying device **1470** may appropriately convert power supplied from an external power source and/or a battery (not shown) embedded in the system **1000** to supply the power to each element of the system **1000**.

[0084] The connecting interface **1480** may provide connection between the system **1000** and an external device connected to the system **1000** to exchange data with the system **1000**. The connecting interface **1480** may be implemented in a variety of interface ways such as an Advanced Technology Attachment (ATA), Serial ATA (SATA), external SATA (e-SATA), Small Computer Small Interface (SCSI), Serial Attached SCSI (SAS), Peripheral Component Interconnection (PCI), PCI express (PCIe), NVM express (NVMe), IEEE 1394, universal serial bus (USB), Secure Digital (SD) card, Multi-Media Card (MMC), embedded multi-media card (eMMC), Universal Flash Storage (UFS), embedded Universal Flash Storage (eUFS) and Compact Flash (CF) card interface.

[0085] FIG. **11** is an exemplary block diagram illustrating a data center to which a storage device according to some embodiments is applied.

[0086] Referring to FIG. **11**, a data center **2000** is a facility for providing a service by collecting various data, and may be referred to as a data storage center. The data center **2000** may be a system for a search engine or a database operation, and may be a computing system used in an enterprise such as a bank or a government agency. The data center **2000** may include application servers **2100_1** to **2100_***n* and storage servers **2200_1** to **2200_***m*. The number of application servers **2100_1** to **2100_***n* and the number of storage servers **2200_1** to **2200_***m* may be variously selected in accordance with embodiments, and the number of application servers **2100_1** to **2100_***n* and the number of storage servers **2200_1** to **2200_***m* may be different from each other.

[0087] The application server **2100_***n* or the storage server **2200_***m* may include at least one of the processors **2110_***n* and **2210_***m* or the memories **2120_***n* and **2220_***m*. The storage server **2200_***m* will be described by way of example. The processor **2210_***m* may control the overall operation of the storage server **2200_***m*, and may access the memory **2220_***m* to execute command languages and/or data loaded into the memory **2220_***m*. The memory **2220_***m* may be a Double Data Rate Synchronous DRAM (DDR SDRAM), a High Bandwidth Memory (HBM), a Hybrid Memory Cube (HMC), a Dual In-line Memory Module (DIMM), an Optane DIMM and/or a Non-Volatile DIMM (NVMDIMM). In accordance with the embodiment, the number of processors **2210_***m* and the number of memories **2220_***m*, which are included in the storage server **2200_***m*, may be variously selected. In one embodiment, the processor **2210_***m* and the memory **2220_***m* may provide a processor-memory pair. In one embodiment, the number of processors **2210_***m* and the number of memories **2220_***m* may be different from each other. The processor **2210_***m* may include a single core processor or a multi-core processor. The description of the storage server **2200_***m* may be similarly applied to the application server **2100_***n*. In accordance with the embodiment, the application server **2100_***n* may not include the storage device **2150_***n*. The storage server **2200_***m* may include at least one storage device **2250_***m*. The number of storage devices **2250_***m* included in the storage server **2200_***m* may be variously selected in accordance with the embodiments.

[0088] Although not shown, the storage device **2250_***m* may include the host interface described with reference to FIGS. **1** to **11**.

[0089] The application servers **2100_1** to **2100_***n* and the storage servers **2200_1** to **2200_***m* may perform communication with each other through a network **3300**. The network **3300** may be implemented using a Fiber Channel (FC) or Ethernet. In this case, the FC is a medium used for relatively high-speed data transmission, and may use an optical switch that provides high performance/high availability. In accordance with an access scheme of the network **3300**, the storage servers **2200_1** to **2200_***m* may be provided as file storages, block storages or object storages.

[0090] In one embodiment, the network **3300** may be a storage-only network such as a storage area network (SAN). For example, the SAN may be an FC-SAN that uses an FC network and is implemented in accordance with an FC protocol (FCP). For another example, the SAN may be an IP-SAN that uses a TCP/IP network and is implemented in accordance with an SCSI over TCP/IP or Internet SCSI (iSCSI) protocol. In another embodiment, the network **3300** may be a general network such as a TCP/IP network. For example, the network **3300** may be implemented in accordance with protocols such as FC over Ethernet (FCoE), Network Attached Storage (NAS) and NVMe over Fabrics (NVMe-oF).

[0091] Hereinafter, the description will be based on the application server **2100**_*n* and the storage server **2200**_*m*. The description of the application server **2100**_*n* may be applied to other application server **2100**_*n*, and the description of the storage server **2200**_*m* may be applied to other storage server **2200**_*m*.

[0092] The application server **2100**_*n* may store data requested by a user or a client in one of the storage servers **2200_1** to **2200**_*m* through the network **3300**. Also, the application server **2100** may acquire the data requested by the user or the client from one of the storage servers **2200_1** to **2200**_*m* through the network **2300**. For example, the application server **2100**_*n* may be implemented as a web server or a database management system (DBMS).

[0093] The application server **2100**_*n* may access the memory **2120**_*n* or the storage device **2150**_*n*, which is included in other application server **2100**_*n*, through the network **3300**. Alternatively, the application server **2100**_*n* may access the memories **2220_1** to **2220**_*m* or the storage devices **2250_1** to **2250**_*m*, which are included in the storage servers **2200_1** to **2200**_*m*, through the network **3300**. Therefore, the application server **2100**_*n* may perform various operations for the data stored in the application servers **2100_1** to **2100**_*n* and/or the storage servers **2200_1** to **2200**_*m*. For example, the application server **2100**_*n* may execute command languages for moving or copying data between the application servers **2100_1** to **2100**_*n* and/or the storage servers **2200_1** to **2200**_*m*. In this case, the data may be moved from the storage devices **2250_1** to **2250**_*m* of the storage servers **2200_1** to **2200**_*m* to the memories **2220_1** to **2220**_*m* of the storage servers **2200_1** to **2200**_*m*, or may be directly moved to the memories **2120_1** to **2120**_*n* of the application servers **2100_1** to **2100**_*n*. The data moved through the network **3300** may be data encrypted for security or privacy.

[0094] The storage server **2200**_*m* will be described by way of example. The interface **2254**_*m* may provide physical connection of the processor **2210**_*m* and a controller **2251**_*m* and physical connection of a Network InterConnect (NIC) **2240**_*m* and the controller **2251**_*m*. For example, the interface **2254**_*m* may be implemented in a Direct Attached Storage (DAS) scheme that directly connects the storage device **2250**_*m* to a dedicated cable. Also, for example, the interface **2254**_*m* may be implemented in the variety of interface ways such as an Advanced Technology Attachment (ATA), Serial ATA (SATA), external SATA (e-SATA), Small Computer Small Interface (SCSI), Serial Attached SCSI (SAS), Peripheral Component Interconnection (PCI), PCI express (PCIe), NVM express (NVMe), IEEE 1394, universal serial bus (USB), Secure Digital (SD) card, Multi-Media Card (MMC), embedded multi-media card (eMMC), Universal Flash Storage (UFS), embedded Universal Flash Storage (eUFS) and Compact Flash (CF) card interface.

[0095] The storage server **2200**_*m* may further include a switch **2230**_*m* and an NIC **2240**_*m*. The switch **2230**_*m* may selectively connect the processor **2210**_*m* with the storage device **2250**_*m* in accordance with the control of the processor **2210**_*m*, or may selectively connect the NIC **2240**_*m* with the storage device **2250**_*m*.

[0096] In one embodiment, the NIC **2240**_*m* may include a network interface card, a network adapter and the like. The NIC **2240**_*m* may be connected to the network **3300** by a wired interface, a wireless interface, a Bluetooth interface, an optical interface and the like. The NIC **2240**_*m* may include an internal memory, a Digital Signal Processor (DSP), a host bus interface and the like, and may be connected to the processor **2210**_*m* and/or the switch **2230**_*m* through the host bus interface. The host bus interface may be implemented as one of the above-described examples of the interface **2254**_*m*. In one embodiment, the NIC **2240**_*m* may be integrated with at least one of the processor **2210**_*m*, the switch **2230**_*m* or the storage device **2250**_*m*.

[0097] In the storage servers **2200_1** to **2200**_*m* or the application servers **2100_1** to **2100**_*n*, the processor may transmit a command to the storage devices **2150_1** to **2150**_*n* and **2250_1** to **2250**_*m* or the memories **2120_1** to **2120**_*n* and **2220_1** to **2220**_*m* to program or read data. At this time, the data may be error-corrected data through an Error Correction Code (ECC) engine. The data may be data processed with Data Bus Inversion (DBI) or Data Masking (DM), and may include Cyclic Redundancy Code (CRC) information. The data may be data encrypted for security or privacy.

[0098] The storage devices **2150_1** to **2150**_*n* and **2250_1** to **2250**_*m* may transmit a control signal and command/address signals to NAND flash memory devices **2252_1** to **2252**_*m* in response to a read command received from the processor. Therefore, when reading data from the NAND flash memory devices **2252_1** to **2252**_*m*, a Read Enable (RE) signal may be input as a data output control signal to output the data to a DQ bus. A data strobe DQS may be generated using the RE signal. The command and the address signal may be latched into a page buffer in accordance with a rising edge or a falling edge of a write enable (WE) signal.

[0099] The controller **2251**_*m* may generally control the operation of the storage device **2250**. In one embodiment, the controller **2251**_*m* may include a Static Random Access Memory (SRAM). The controller **2251**_*m* may write data in the NAND flash **2252**_*m* in response to a write command, or may read data from the NAND flash **2252**_*m* in response to a read command. For example, the write command and/or the read command may be provided from the processor **2210**_*m* in the storage server **2200**_*m*, the processor **2210**_*m* in the other storage server **2200**_*m* or the processors **2110_1** and **2110**_*n* in the application servers **2100_1** and **2100**_*n*. The DRAM **2253**_*m* may temporarily store (buffer) data to be written in the NAND flash **2252**_*m* or data read from the NAND flash **2252**_*m*. Also, the DRAM **2253**_*m* may store metadata. In this case, the metadata is user data or data generated by the controller **2251**_*m* to manage the NAND flash **2252**_*m*. The storage device **2250**_*m* may include a Secure Element (SE) for security or privacy.

[0100] Although the embodiments of the present disclosure have been described with reference to the accompanying drawings, it will be apparent to those skilled in the art that the present disclosure may be embodied in other specific forms without departing from technical spirits and essential characteristics of the present disclosure. Thus, the above embodiments are to be considered in all respects as illustrative and not restrictive.

Claims

1. A storage device comprising: a storage controller comprising an address translation cache that translates a virtual address received with a command from a host into a physical address; and a non-volatile memory device, wherein the storage controller stores index information on a host Specific Memory Buffer (SMB) in the host in advance, performs direct memory access to the virtual address for the host SMB when a cache miss occurs in the address translation cache, and transmits an exceptional Completion Queue (CQ) including an indicator corresponding to a location where the direct memory access is performed to the host when the direct memory access is completely performed.

2. The storage device of claim 1, wherein the storage controller comprises: a specific memory buffer (SMB) manager requesting memory access to the virtual address when the cache miss occurs; a direct memory access manager performing an access operation for the host SMB together with the virtual address upon receiving the memory access request from the SMB manager; a device SMB mapping the virtual address to an indicator of the index information and storing the mapped address upon the memory access request of the direct memory access manager; and a device completion manager transmitting the exceptional completion queue when the access operation of the direct memory access manager is completed.

3. The storage device of claim 1, wherein the index information comprises physical addresses of the host SMB, a block size, and an index number corresponding to the indicator.

4. The storage device of claim 1, wherein the host comprises a host processor and a host memory, and the host memory comprises a host completion manager receiving the completion queue from the storage controller in accordance with direct memory access operation, and the host SMB storing the index information shared in advance in the storage controller and storing data accessed

via direct memory access together with the virtual address.
5. The storage device of claim 4, wherein the host maps the virtual address received through the direct memory access operation to the physical address corresponding to the indicator included in the completion queue and stores the mapped address as address translation information.
6. The storage device of claim 1, wherein the storage controller transmits an address translation service request to the host when a cache miss occurs in the address translation cache, updates the physical address allocated to the virtual address in the address translation cache when a physical address allocated in accordance with the address translation service request is received from the host (Page Hit), and performs direct memory access to the host SMB of the allocated physical address.
7. A storage system comprising: a host comprising a host memory storing index information on a physical address and a host processor outputting a command and a virtual address; and a storage device sharing the index information in advance and transmitting data according to the command to the host memory together with an indicator according to the index information in accordance with a direct memory access operation.
8. The storage system of claim 7, wherein the host memory comprises: a data buffer storing operation data; a host completion manager receiving a completion queue, which comprises the indicator, from the storage controller in accordance with the direct memory access operation; and a host SMB storing the index information shared in advance in the storage device.
9. The storage system of claim 8, wherein the host further comprises a translation agent mapping the virtual address for the direct memory access operation to the physical address corresponding to the indicator to update the mapped address with address translation information.
10. The storage system of claim 9, wherein the storage device comprises: an address translation cache translating the virtual address to the physical address of the host SMB; a specific memory buffer (SMB) manager requesting memory access to the virtual address when a cache miss occurs in the address translation cache; a direct memory access manager performing an access operation for the host SMB together with the virtual address upon receiving the memory access request from the SMB manager; a device SMB storing the index information shared in advance; and a device completion manager transmitting a completion queue, which comprises the indicator, when the access operation of the direct memory access manager is completed.
11. The storage system of claim 10, wherein the device completion manager allocates any one indicator for a location in the host SMB to the virtual address based on the stored index information so that the indicator is included in the completion queue.
12. The storage system of claim 10, wherein the address translation cache requests the translation agent of an address translation service when the cache miss occurs due to no physical address corresponding to the virtual address, and the SMB manager requests the direct memory access manager of the memory access when a newly allocated physical address is not received from the translation agent (Page Miss).
13. The storage system of claim 12, wherein the address translation cache maps the received physical address to the virtual address and updates the mapped physical address when a newly allocated physical address is received from the translation agent (Page Hit), and the direct memory access manager performs direct memory access to the updated physical address of the host SMB.
14. The storage system of claim 13, wherein the device completion manager transmits a normal completion queue to a host completion manager after performing the direct memory access to the updated physical address of the host SMB.
15. The storage system of claim 8, wherein the index information comprises physical addresses of the host SMB, a block size, and an index number corresponding to the indicator.
16. An operating method of a storage device, the operating method comprising: performing direct memory access to a host together with a virtual address when a cache miss for the virtual address occurs in an address translation cache; and outputting a completion queue, which comprises an

indicator corresponding to the virtual address, to the host.

17. The operating method of claim 16, wherein the storage device receives and stores index information on a host memory from the host in advance.

18. The operating method of claim 17, wherein the storage device selects any one indicator for a location in the host memory based on the index information with respect to the cache missed virtual address, and outputs the selected indicator to the host by including the selected indicator in the completion queue. Page **6**

19. The operating method of claim 18, wherein the host maps a physical address corresponding to the indicator, received from the storage device, of the index information to the virtual address, and performs the direct memory access to a location of the mapped physical address in the host memory.

20. The operating method of claim 19, wherein the host transmits the physical address mapped to the virtual address to the address translation cache through a translation agent.

21. (canceled)

22. (canceled)
